# Abhijit Shankar Mudigonda


**Contact Information**

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https://abhijit-mudigonda.github.io/math


**Research Interests**

Number Theory, Algebraic Geometry, Theoretical Computer Science

**Education**

**University of Chicago**

Ph.D. in Mathematics and Computer Science, September 2021 - June 2027
Advised by Frank Calegari and Janos Simon
M.S. in Computer Science, June 2024

**Massachusetts Institute of Technology**

B.S. in Mathematics, September 2015 - June 2019

**Publications**

A. S. Mudigonda. *Computing Hilbert modular forms with nonparitious weight*. Research in Number Theory, 2026. Proceedings of the Algorithmic Number Theory Symposium XVII, to appear.

B. Matschke and A. S. Mudigonda. *Quadratic Fields Admitting Elliptic Curves with Rational j-Invariant and Good Reduction Everywhere*. Journal of Number Theory. Volume 247, June 2023, Pages 162-210.

A. S. Mudigonda and R. R. Williams. *Time-Space Lower Bounds for Simulating Proof Systems with Quantum and Randomized Verifiers*. Proceedings of the 2021 Conference on Innovations in Theoretical Computer Science.

**Preprints**

A. S. Mudigonda. *Potential modularity of some atypical abelian fourfolds*.

E. Assaf, A. S. Mudigonda. *Computing Hilbert modular forms with nontrivial nebentypus*.

E. Assaf, A. Babei, B. Breen, S. Chari, E. Costa, J. Duque-Rosero, A. Horawa, J. Kieffer, A. Kulkarni, A. S. Mudigonda, M. Musty, S. Schiavone, S. Sethi, S. Tripp, J. Voight. *Computing with Fourier expansions of Hilbert modular forms*. In preparation.

**Patents**

D. Gutfreund, Q. Fan, A. Mudigonda. *Restructuring Deep Neural Networks to Reduce the Number of Parameters*. USPA #20200160144. Published May 21, 2020.

**Invited Talks**

January 2027 – JMM Special Session, Explicit Methods in Number Theory
January 2027 – JMM Special Session, Mathematics Informed by Computing
December 2026 – UW-Madison Number Theory Seminar
November 2026 – McGill Graduate Student Number Theory Seminar
October 2026 – University of Sydney Computational Algebra Seminar
July 2026 – Algorithmic Number Theory Symposium XVII
May 2026 – UChicago Number Theory Seminar
April 2026 – UC Santa Barbara Number Theory Seminar
January 2021 – Innovations in Theoretical Computer Science
January 2021 – UT Austin Quantum Information Seminar

## Work Experience

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|---|---|
| 08/2020 - 09/2021 | *Software Engineer, Facebook Reality Labs*<br>OS engineer on the XROS platform team, developing features for the Oculus Quest Pro. |
| 01/2017 - 04/2017 | *Research Intern, IBM Research.*<br>Developed neural network architectures inspired by expander graphs. |
| 06/2016 - 08/2016 | *Research Intern, Knight Cancer Institute.*<br>Used deep learning models to predict DNA sequencing accuracy. |

## Teaching Experience

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|---|---|
| 09/2022- Present | *Teaching Assistant*<br>Served as a TA for Honors Introduction to Discrete Mathematics, Introduction to Algorithms, Honors Introduction to Complexity Theory, Algebra I, Introduction to Riemannian Geometry, and Introduction to Algebraic Number Theory |
| 09/2022 - 06/2023 | *UChicago Teaching Fellow*<br>Underwent a year-long training program preparing graduate students to be independent lecturers. |
| 06/2020 - 08/2020 | *Summer STEM Institute Research Mentor*<br>Mentored two high school students in theoretical and applied quantum computing research projects. |
| 01/2019 | *MIT Global Teaching Labs Teacher*<br>Taught physics and biology to high school students at the Liceo Scientifico Michelangelo Grigoletti in Pordenone, Italy. |
| 06/2015 - 06/2018 | *USA Biology Olympiad Teaching Assistant*<br>Wrote practical and theoretical examinations to select the 2015, 2016, 2017, and 2018 U.S. representative teams.<br>Gave lectures on topics ranging from developmental physiology to biostatistics. |

## Refereeing

Algebra & Number Theory

## Honors and Awards

2021 NSF Graduate Research Fellow
2021 UChicago Crerar Fellow
2014 International Biology Olympiad - Silver Medal ($25^{\text{th}}$)
2015 North American Computational Linguistics Olympiad - U.S. Alternate ($11^{\text{th}}$)